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Course Code: CSA0920RC

Course Name: Programming in Java

Slot: C

CO2_Assignment - 1

Q2. Length, Substring & Index Operations Write a Java program that accepts a sentence or word from the user and performs three separate String operations on it: determine its total length using `length()`, extract a specific portion of the string using `substring()` (e.g. the first five characters, or characters between two given positions), and locate the position of a specific character or substring within it using `indexOf()`. Display the results of each operation clearly, labeled by which method produced it.

Concepts: `length()`, `substring()`, `indexOf()`.

CODE:

```
import java.util.Scanner;
public class java2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a sentence or word: ");
        String str = sc.nextLine();
        System.out.println("\nUsing length()");
        System.out.println("Total Length = " + str.length());
        System.out.println("\nUsing substring()");
        if (str.length() >= 5) {
            System.out.println("First 5 characters = " + str.substring(0, 5));
        } else {
            System.out.println("String has less than 5 characters.");
        }
        System.out.print("\nEnter a character or substring to search: ");
        String search = sc.nextLine();
```

```

int position = str.indexOf(search);

System.out.println("\nUsing indexOf():");
if (position != -1) {
    System.out.println "\"" + search + "\" found at index " + position);
} else {
    System.out.println "\"" + search + "\" not found.");
}

sc.close();
}
}

```

OUTPUT:

```

Run java2 x
"C:\Program Files\Java\jdk-17.0.18\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\lib\idea_rt...
Enter a sentence or word: java

Using length():
Total Length = 4

Using substring():
String has less than 5 characters.

Enter a character or substring to search: a

Using indexOf():
"a" found at index 1

Process finished with exit code 0

```